

POD PDF Projection and the CMS Inclusive-Jet Fit

Inclusive Jets / xFitter working note

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1 POD representation and projection

The POD construction follows the linear PDF-model idea of Ref. [3]. At $Q_0 = 1.65 \text{ GeV}$, LHAPDF member zero of 250503_pod_basis_40k is $f_a^{(0)}$, while the other 100 members define shifts $B_{ai} = f_a^{(i)} - f_a^{(0)}$ for $a = (g, u, \bar{u}, d, \bar{d}, s, \bar{s}, c, \bar{c}, b, \bar{b})$. All quantities are xf_a . The POD input PDF is

$$f_a^{\text{POD}}(x, Q_0; \mathbf{c}) = f_a^{(0)}(x, Q_0) + \sum_{i=1}^{100} B_{ai}(x) c_i. \quad (1)$$

QCDNUM evolves the PDF, not the coefficients [5]. For an external target t sampled at the same scale and grid, let $d = t - f^{(0)}$ and form the matrix X from the shifts. The generic projection is

$$\hat{c} = \arg \min_{\mathbf{c}} (X\mathbf{c} - d)^T W (X\mathbf{c} - d), \quad (X^T W X) \hat{c} = X^T W d. \quad (2)$$

This is a PDF-space calculation, not a data-likelihood fit. The generic CT18NNLO validation uses native shifts without amplitude rescaling and an absolute metric. A target at Q_{ext} must be projected against the basis evaluated at the same Q_{ext} .

2 CMS Figure–2 modifications

The retained CMS objective supplements the absolute term by relative residual penalties,

$$\chi_{\text{proj}}^2 = \|Xc - d\|_2^2 + \lambda_g^2 \sum_{0.05 < x < 0.99} \left(\frac{r_g}{\max(|t_g|, \epsilon)} \right)^2 \quad (3)$$

$$+ \lambda_v^2 \sum_{10^{-4} < x < 0.1} \sum_{q=u,d} \left(\frac{r_{q-\bar{q}}}{\max(|t_{q-\bar{q}}|, \epsilon)} \right)^2 + \lambda_{F_2}^2 \sum_{10^{-4} < x < 0.1} \left(\frac{r_{F_2}}{\max(|t_{F_2}|, \epsilon)} \right)^2. \quad (4)$$

Here r is target minus reconstruction and $F_2^{\text{proxy}} = \sum_{q=u,d,s,c,b} e_q^2 (q + \bar{q})$. The production settings are $\lambda_g = 0.1$, $\lambda_v = 0.1$, and $\lambda_{F_2} = 3$. These are projection objectives only, not experimental likelihood terms or PDF priors.

The relative high- x gluon term is the *gluon-tail enhancement*. An absolute norm otherwise underweights a small but phenomenologically relevant high- x gluon. The valence and F_2 terms protect low- x DIS-relevant light-quark combinations.

The direct HERAPDF input starts at $\sqrt{1.9}$ GeV while POD starts at 1.65 GeV. Valid closure therefore exports the direct PDF on a full LHAPDF grid down to $x = 10^{-9}$, reimports it at 1.65 GeV, and evolves both inputs through the same QCDNUM route. This prevents grid, input-scale, and evolution-history effects from being called POD compression. The implementation is in xFitter [4], with the CMS/HERA configuration supplied alongside the CMS analysis [1, 2].

Pointwise high- x $u, \bar{u}, d, \bar{d}$ relative terms were rejected because vanishing antiquarks are ill conditioned. Stable high- x valence/ F_2 trials also did not improve closure. The full basis spans the light combinations but only along a highly ill-conditioned direction that catastrophically damages gluon closure; these trials are diagnostics, not production settings.

3 Direct CMS analytic parametrisation

At $Q_0^2 = 1.9 \text{ GeV}^2$, the direct Figure-2 fit uses $xf = Ax^B(1-x)^C(1+Dx+Ex^2)$, specifically

$$xg = A_g x^{B_g} (1-x)^{C_g} (1+D_g x + E_g x^2), \quad x u_v = A_{u_v} x^{B_{u_v}} (1-x)^{C_{u_v}} (1+E_{u_v} x^2), \quad (5)$$

$$x d_v = A_{d_v} x^{B_{d_v}} (1-x)^{C_{d_v}}, \quad x \bar{U} = A_{\bar{U}} x^{B_{\bar{D}}} (1-x)^{C_{\bar{U}}} (1+D_{\bar{U}} x), \quad (6)$$

$$x \bar{D} = A_{\bar{D}} x^{B_{\bar{D}}} (1-x)^{C_{\bar{D}}} (1+E_{\bar{D}} x^2). \quad (7)$$

The conventions are $\bar{U} = \bar{u}$, $\bar{D} = \bar{d} + \bar{s}$, $x\bar{d} = (1-f_s)x\bar{D}$, $x\bar{s} = f_s x\bar{D}$, $f_s = 0.4$, and $A_{\bar{U}} = (1-f_s)A_{\bar{D}}$. Valence-number and momentum sum rules fix A_g, A_{u_v}, A_{d_v} . The 16 independent parameters are $B_g, C_g, D_g, E_g, B_{u_v}, C_{u_v}, E_{u_v}, B_{d_v}, C_{d_v}, A_{\bar{D}}, B_{\bar{D}}, C_{\bar{D}}, E_{\bar{D}}, D_{\bar{U}}, E_{\bar{U}}$.

4 Other global-fit parametrisations

4.1 CT18NNLO

CT18NNLO uses the CTEQ-TEA Hessian framework and low-scale endpoint powers times Bernstein-polynomial expansions [6], $xf_a = x^{a_{1,a}}(1-x)^{a_{2,a}}P_a(y)$, where y is a monotonic map such as $\sqrt{x}$. Bernstein polynomials give local shape control; sum rules and flavour relations apply at Q_0 ; Hessian eigenvectors encode uncertainties. CT18NNLO is the target used by the generic POD validation.

4.2 MSHT and NNPDF

MSHT uses endpoint powers times flexible Chebyshev-polynomial expansions with Hessian uncertainties [7]. NNPDF uses neural networks with pre-processing factors $xf_a = x^{\alpha_a}(1-x)^{\beta_a}\text{NN}_a(x)$ and Monte-Carlo replicas [8]. All are input-scale parametrisations followed by DGLAP evolution. POD instead fixes shape functions from an ensemble and fits only linear coordinates.

5 Scope

The scale-aligned full 100-mode POD has small local likelihood differences from the direct HERA+CMS fit. This does not validate a five- or seven-mode fit; every truncation requires its own closure and minimised-fit validation.

References

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